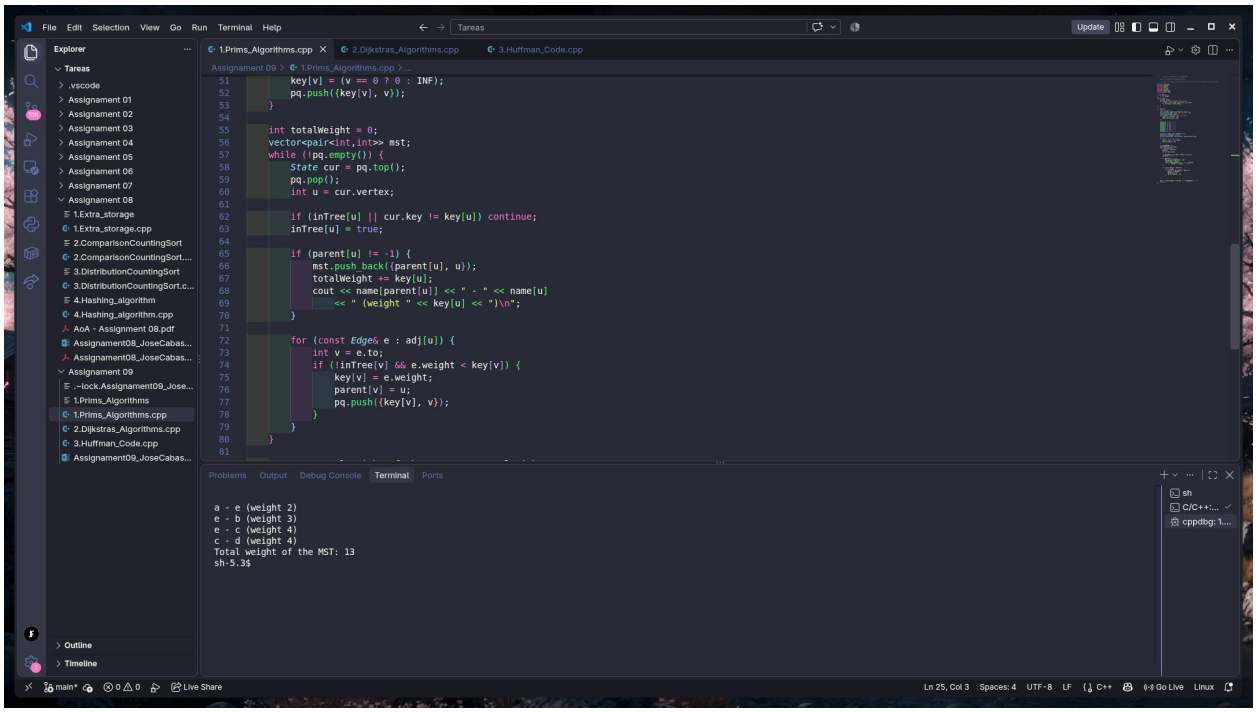
**Analysis of Algoritms**

**Assignament 09**

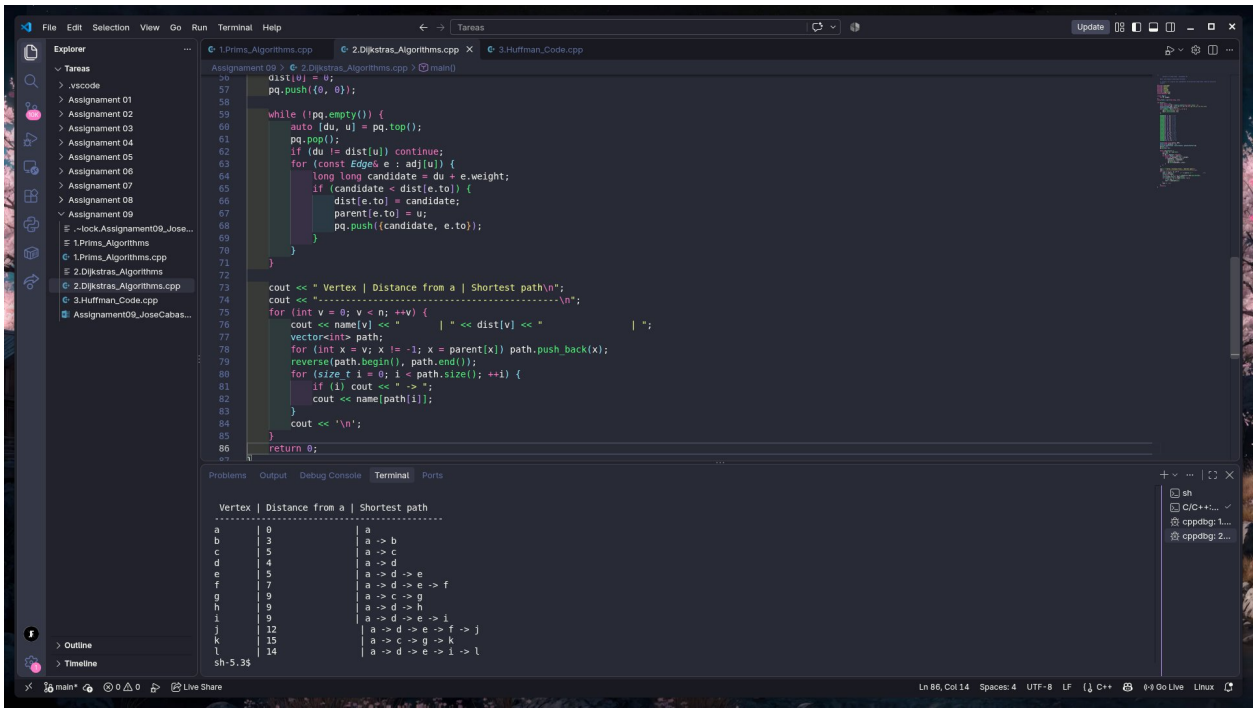
**Name:** Jose Ignacio Cabascango Fernandez

**Link GitHub:** https://github.com/CodeWithJose1/Analysis-of-Algorithms/tree/main/Assignament-09/src

## 2. Create a C++ program that implementes the Prim's Algorithms. Show an execution example.




## 4. Create a C++ program that implementes the Dijkstra's Algorithms. Show and execution example.

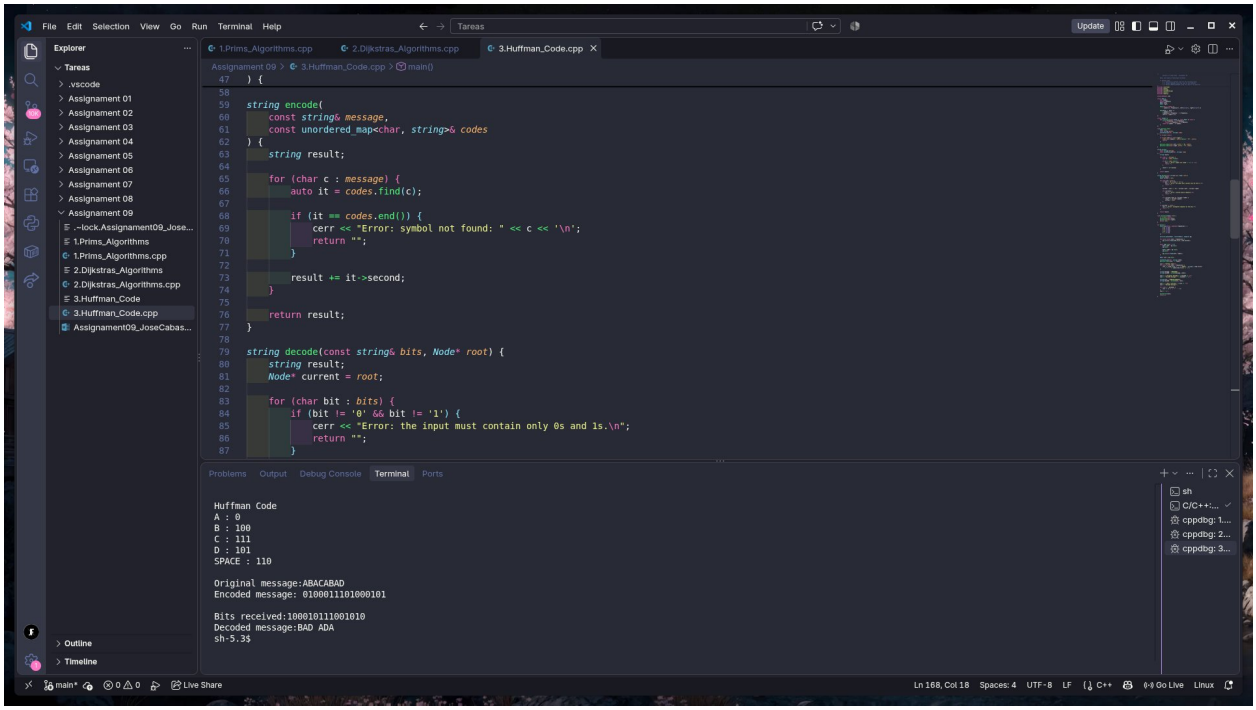


## 5. Huffman Code.

- **5.1. Construct a Huffman code for the following data:**

| symbol | A | B | C | D | _ |
|---|---|---|---|---|---|
| frequency | 0.4 | 0.1 | 0.2 | 0.15 | 0.15 |

- **5.2. Encode ABACABAD using the code of the question**
- **5.3. Decode 100010111001010 using the code of the question**

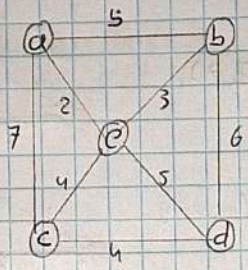


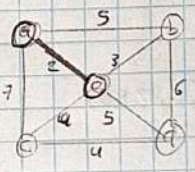
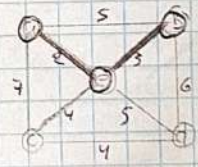
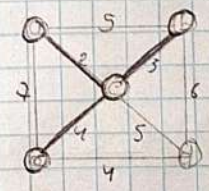
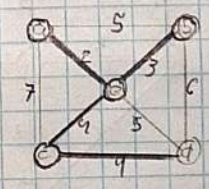
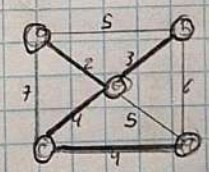
# Analysis of Algorithms

## Assignament 09

Name: Jose Ignacio Cabezuelo Fernández

1. Apply Prim's algorithm to the following graph. Include in the priority queue all the vertices not already in the tree

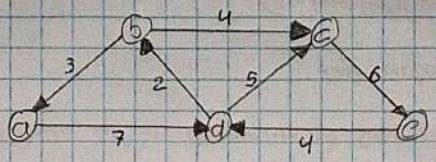


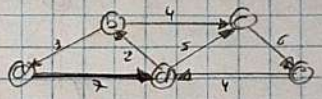
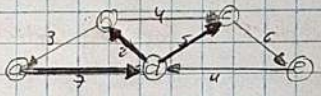
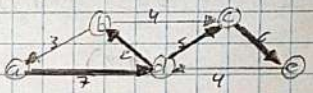
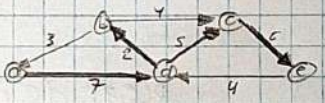

| Tree vertices | Remaining vertices | Illustration |
|---|---|---|
| a(–, –) | b(a,5), c(a,7), d(–,∞) |  |
| e(a,2) | b(e,3), c(e,4), d(e,5) |  |
| b(e,3) | c(e,4), d(e,5) |  |
| c(e,4) | d(c,4) |  |
| d(c,4) | |  |

Selected Edges: a-e(2), e-b(3), e-c(4), c-d(4); Total weight = 2+3+4+4 = 13.

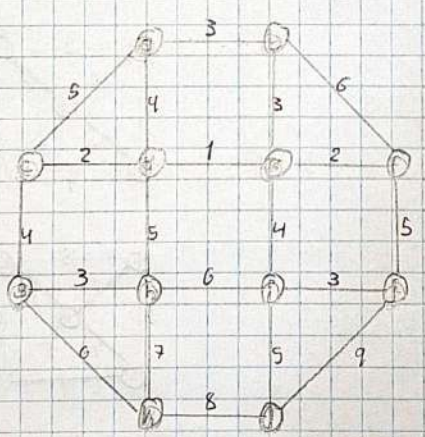
3. Solve (paper and pencil) the following instances of the single-source shortest-paths problem with vertex a as the source using Dijkstra's Algorithm:

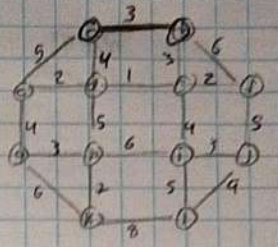
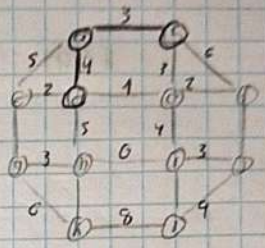
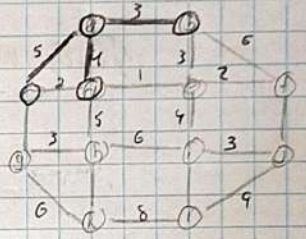
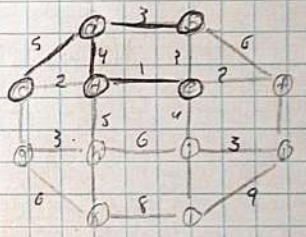
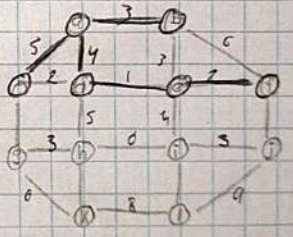
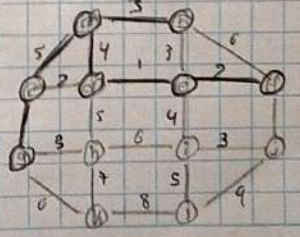
a.

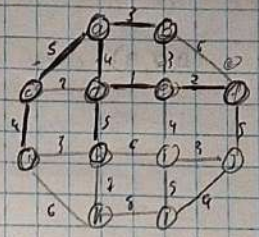
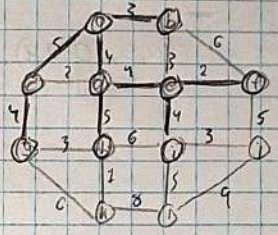
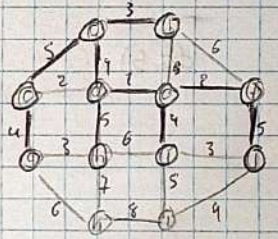
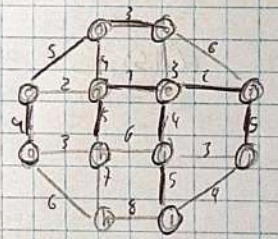
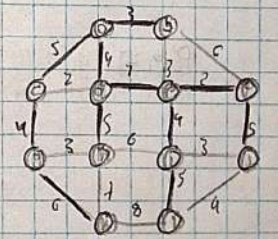
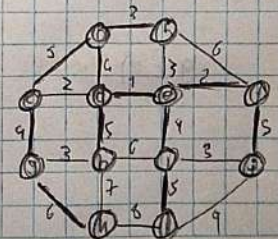



| Tree vertices | Remaining vertices | Illustration |
|---|---|---|
| a(-, 0) | b(a, 3), c(-, ∞), d(a, 7), e(-, ∞) |  |
| d(a, 7) | c(d, 7+5), b(d, 7+2), e(-, ∞) |  |
| b(d, 9) | c(d, 12), e(-, ∞) |  |
| d(d, 12) | e(c, 12+6) |  |
| c(c, 18) | |  |

b.



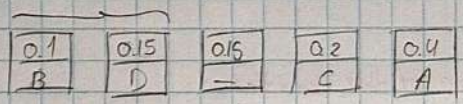
| Tree vertices | Remaining vertices | Illustration |
| --- | --- | --- |
| a(-,0) | b(a,3), c(a,5), d(a,4), e(-,∞), f(-,∞) g(-,∞), h(-,∞), i(-,∞), j(-,∞), k(-,∞) l(-,∞) |  |
| b(a,3) | c(a,5), d(a,4), e(b,6), f(b,9), g(-,∞) h(-,∞) i(-,∞) j(-,∞) k(-,∞) l(-,∞) |  |
| d(a,4) | d(a,5), e(d,5), f(b,9), g(-,∞), h(d,9) i(-,∞), j(-,∞), k(-,∞), l(-,∞) |  |
| c(a,5) | e(d,5), f(b,9), g(c,9), h(d,9), i(-,∞) j(-,∞), k(-,∞), l(-,∞) |  |
| e(d,5) | f(e,7), g(c,9), h(d,9), i(e,9), j(-,∞) k(-,∞), l(-,∞) |  |
| f(e,7) | g(c,9), h(d,9), i(e,9), j(f,12), k(-,∞) l(-,∞) |  |

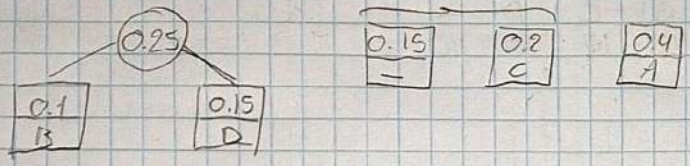
| Tree Vertices | Remaining Vertices | Illustration |
| --- | --- | --- |
| g(c, 9) | h(d, 9), i(e, 9), j(f, 12), k(g, 15), l(-, ∞) |  |
| h(d, 9) | i(e, 9), j(f, 12), k(g, 15), l(-, ∞) |  |
| i(e, 9) | j(f, 12), k(g, 15), l(i, 14) |  |
| j(f, 12) | k(g, 15), l(i, 14) |  |
| l(i, 14) | k(g, 15) |  |
| k(g, 15) | |  |

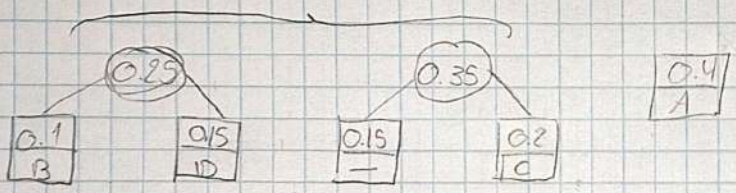
5. Huffman Code

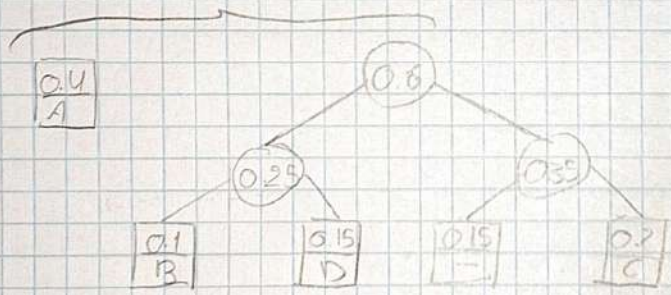
5.1 Construct a Huffman Code for the following data:

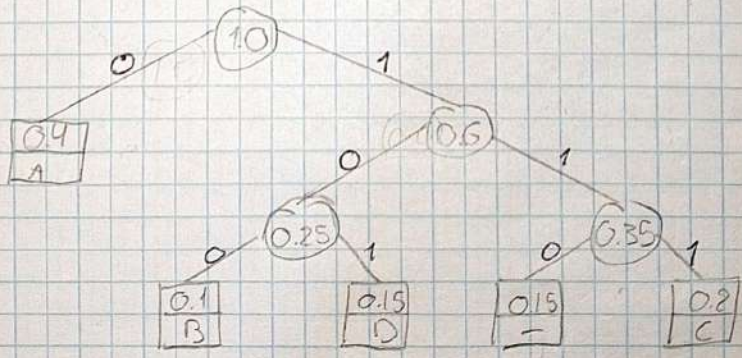
| symbol | A | B | C | D | — |
|---|---|---|---|---|---|
| frequency | 0.4 | 0.1 | 0.2 | 0.15 | 0.15 |




| Symbol | A | B | C | D | — |
|---|---|---|---|---|---|
| frequency | 0.4 | 0.1 | 0.2 | 0.15 | 0.15 |
| codeword | 0 | 100 | 111 | 101 | 110 |

5.2 Encode ABACABAD using the code of the question.

| Symbol | A | B | C | D | _ |
|---|---|---|---|---|---|
| codeword | 0 | 100 | 111 | 101 | 110 |

Encoded string : 0100011101000101. 16 bits.

5.3 Decode 1000101111001010 using the code of the question.

| Symbol | A | B | C | D | _ |
|---|---|---|---|---|---|
| codeword | 0 | 100 | 111 | 101 | 110 |

1000101111001010
B A D _ A D A

Decode string : BAD_ADA